SMART GRID TECHNOLOGIES: FINAL COURSEWORK

This exercise is to be completed individually. The answers should be returned as a PDF on Blackboard by Thursday 16 February 2017 (deadline midnight). Submission instructions will be made available. The exercise will be marked and the grade will make up 20% of the final grade for the module; the written exam will make up the remaining 80%.

Excellent answers are characterised by the following:

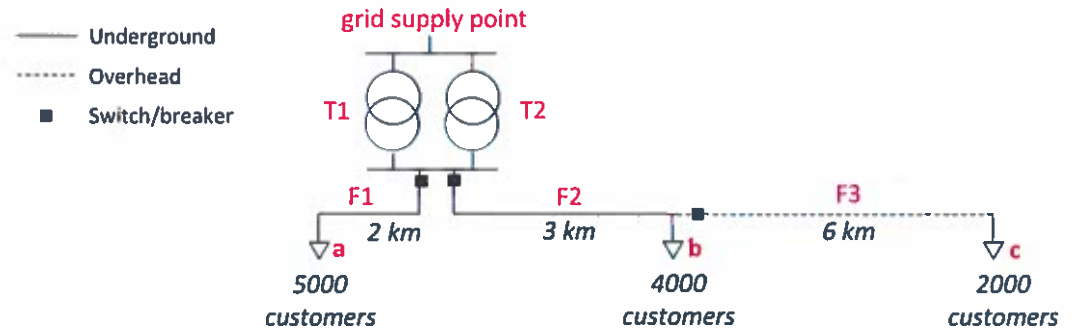
- *Correct results*
- *A clear description of the steps involved in the derivation of those results.*
- *Monte Carlo estimates are accompanied by error estimates.*
- *Careful use of units in calculations and presentation of results.*
- *Inclusion of relevant code in the appendix. This is not a coding class, so the code will not be judged on the use of appropriate/clever programming constructs, but any code must be well-organised and any non-trivial steps must be documented by means of code comments. We assume the use of Matlab, but the use of Python, Mathematica or C/C++ is also acceptable.*
- *No plagiarism: you may discuss the questions with each other, but the exercises must be completed on an individual basis. Any non-original text or code will affect your mark.*
- *Where appropriate, include references to source material, including reports, books and papers.*

Tips:

- *Do not rely only on lecture notes. Use books, other documentation or the internet if required.*
- *Make use of the ability to cross-check analytical and Monte Carlo results. If the complexity of the network is an issue (for analytical and/or Monte Carlo studies), try reducing the size of the network to identify and correct any errors.*

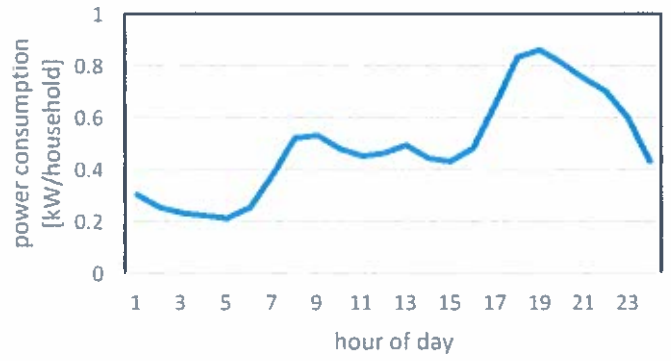
SYSTEM DESCRIPTION

You will analyse the reliability of a simple distribution system using three methods: an analytical approach, snapshot Monte Carlo simulations and time-sequential Monte Carlo simulations. The distribution network is depicted below. The diagram lists the number of customers (residential households) at each load point and the lengths and types of connection. The transformers have a thermal rating of 8MVA each and it may be assumed that the transformer will trip when overloaded. We will consider active power only (assuming unit power factor throughout). The cables have no thermal limits in this exercise.



LOAD MODEL

The residential loads are described by the average daily load profile shown below[1]. Seasonal changes and fluctuations between different days are omitted for this exercise, so the annual profile is a repetition of the daily profile. It is assumed that the load points are fully correlated: the load levels differ only in scale, due to the different number of customers connected to each load point.



[1] Derived from *Elexon profile class 1* (domestic unrestricted customers): https://www.elexon.co.uk/wp-content/uploads/2013/11/load_profiles_v2.0_cgi.pdf

The table below give the average household power consumption for each hour.

Hour of day	Power consumption [kW/household]	Hour of day (continued)	Power consumption [kW/household]
1	0.3	13	0.49
2	0.25	14	0.44
3	0.23	15	0.43
4	0.22	16	0.48
5	0.21	17	0.65
6	0.25	18	0.83
7	0.38	19	0.86
8	0.52	20	0.81
9	0.53	21	0.75
10	0.48	22	0.7
11	0.45	23	0.6
12	0.46	24	0.43

In Matlab, you can use the following command to store the load profile in an array:

```
load_profile = [0.30 0.25 0.23 0.22 0.21 0.25 0.38 0.52 0.53 0.48 0.45
0.46 0.49 0.44 0.43 0.48 0.65 0.83 0.86 0.81 0.75 0.70 0.60 0.43];
```

A load levels in a specific hour can then be selected using `load_profile(hour)`.

RELIABILITY MODEL

The ability to supply power to the households is affected by independent failures in the transformers and overhead/underground cables. Their failure and repair parameters are summarised below.

Component	Fault process	Repair process
Transformer	MTTF = 100 years	MTTR = 10 days
Overhead cable	Failure rate = 0.1/km/year	MTTR = 10 hours
Underground cable	Failure rate = 0.05/km/year	MTTR = 2 days

It is assumed that cable faults are immediately (i.e. within 3 minutes) isolated locally by means of fuses and/or breakers; this has no impact further upstream (i.e. closer to the grid supply point). The affected line section will remain in an isolated state until the repair has taken place. All faults are assumed to be persistent and to last more than 3 minutes, so that all events count for CML/CI calculations. You may assume that the grid supply point is 100% reliable. All component faults are assumed to be independent.

PART 1 – ANALYTICAL CALCULATIONS

1. Compute the availability of each component (T1, T2, F1, …).
2. Compute the loss of load probability (LOLP) for this network. Consider whether a single transformer is sufficient to supply all loads. You may want to use reliability block diagrams.
3. Compute the unavailability of power supplied to each load point (a,b,c). Hint: first calculate the probability that two transformers are required to supply all load, which will depend on the hour of day *and* on whether *other* load points are being supplied.
4. Using the computed availabilities, determine the CML for this network.
5. Would the CML index for this network be considered acceptable for real distribution networks?

PART 2 – STATE SAMPLING MONTE CARLO

Implementing a state sampling Monte Carlo algorithm to estimate a range of reliability indicators: LOLP, expected CML, and EENS.

6. Define what constitutes a 'state' for this problem, and implement a function that samples a network state:
   a. Sample a random state according to the availabilities computed in (1) and the load model defined above. Hint: you can use the Matlab command `randi(.)` to generate a random hour-of-day.
   b. Return the sampled state.
7. Write a function takes a network state as an input and computes three impact values, which will be used to compute the LOLP, expected CML and EENS.
8. Use the functions you created above to develop a state sampling Monte Carlo simulation that estimates the LOLP, expected CML and EENS. Use at least 100,000 samples and report the results accompanied by error estimates.
9. Compare the LOLP and expected CML results to those obtained using analytical computations (question 2 and 4), and reflect on the relative advantages of each method.

PART 3 – TIME-SEQUENTIAL MONTE CARLO

Implementing a time-sequential Monte Carlo algorithm to estimate LOLP, expected CML, EENS and expected CI, alongside their annual distributions.

10. Write a function that samples a random initial state of the system ($t = 0$). The state of the network elements should be sampled according to their availability, and the load can be initialised at hour 1.
11. Determine the discrete time Markov chains for the transformers and feeders in the network, assuming a time step of $\Delta t = 1\ \mathbf{hour}$.
12. Write a function that takes a network state at time t, evolves it to $t + 1\mathbf{hour}$ and returns the new state. Note that the load is *not* modelled as a Markov chain; the load level advances one hour at a time (i.e. exactly 24 hours for a full cycle).
13. Implement a time-sequential MC simulation that generates random paths of length T =1 year. For each year, it should compute the loss-of-load time, CML, CI and ENS. The values for all simulated years can be combined to estimate the LOLP, expected CML, EENS and expected

CI. For the CI index, you can count every single instance customers are disconnected, irrespective of the cause[2].

14. Sample at least 1,000 random years and report estimates (with error estimates) for LOLP, expected CML, EENS and expected CI. Compare the results with those obtained using the analytical method and state sampling simulations.
15. Plot histograms of the annual CML, CI and ENS values. What drives the 'extreme years' with very high CML and CI values? What could a DNO do to specifically reduce the impact in those years? Provide more than one option, and discuss what information you would need to decide to perform a cost-benefit assessment of these options and decide on the best action.

[2] Note that this assumption is technically incorrect, because a single 'event' (e.g. transformer outage) may cause recurrent outages every evening peak. Officially, this should be counted as a single event.